

DraftRun Pipeline AS IS

Current as of Slice 2.17.4.6.1.3.2: DraftRun Validation Critical and Final Gate Warning Repair.

This document is the maintained technical map of the current DraftRun generation pipeline. It describes the running system as it exists now, not the target design. Update this file and regenerate the PDF whenever a slice changes DraftRun steps, artifacts, role model usage, context flow, retry/fallback behavior, validation, ranking, revision, or trace semantics.

PDF quick-view copy: docs/architecture/DRAFT_RUN_PIPELINE_AS_IS.pdf.

Regenerate PDF:

```
python scripts/generate-draft-run-pipeline-pdf.py
```

1. Core concepts

Concept	Meaning	Stored where
DraftRun	Parent orchestration run for one approved post fabula. It owns step order, status, final draft, and child AiRun ids.	var/glavred-draft-runs.sqlite3
DraftRunStep	One logical pipeline stage: context, source intent, public evidence, feasibility, contract, rules, planning, candidates, validation, complete.	draft_run_steps.artifact_payload
AiRun	One model/provider call or audited fallback inside a DraftRun.	var/glavred-ai-runs.sqlite3
SourceLedger	Claim/provenance inventory with allowed use, confidence, warnings, risks, and forbidden inferences.	context and enriched publicEvidence artifacts
EvidenceInterpretation	Editorial interpretation of accepted evidence: implications, tensions, usable examples, limits, forbidden overclaims, reader-value hooks, and rejected uses.	rulePack.evidenceInterpretation, child AiRun trace
PostContract	Locked editorial intent: thesis, audience, CTA, allowed claims, forbidden moves, size contract, topic/fabula obligations.	postContract artifact
RuleRegistrySnapshot	Machine-readable rules and validator bindings derived from contract, ledger, topic, fabula, and publisher rules.	rulePack.ruleRegistrySnapshot
ArticleDossier	DraftRun-local article memory built from artifacts. It is not workspace persistence and not vector storage.	selected step artifacts
ContextPack	Role-specific subset of ArticleDossier passed to LLM calls.	step artifacts and child AiRun.requestPayload
EditorialCritiqueReport	Report-only prosecutor/editor critique of candidate idea strength, tension, author stance, source integration, and reader value.	validation.editorialCritiqueReport, child AiRun trace
AlternativeAngleTournament	Critic-driven challenger route and candidate used to escape a weak local optimum.	validation.alternativeAngleTournament, child AiRun trace

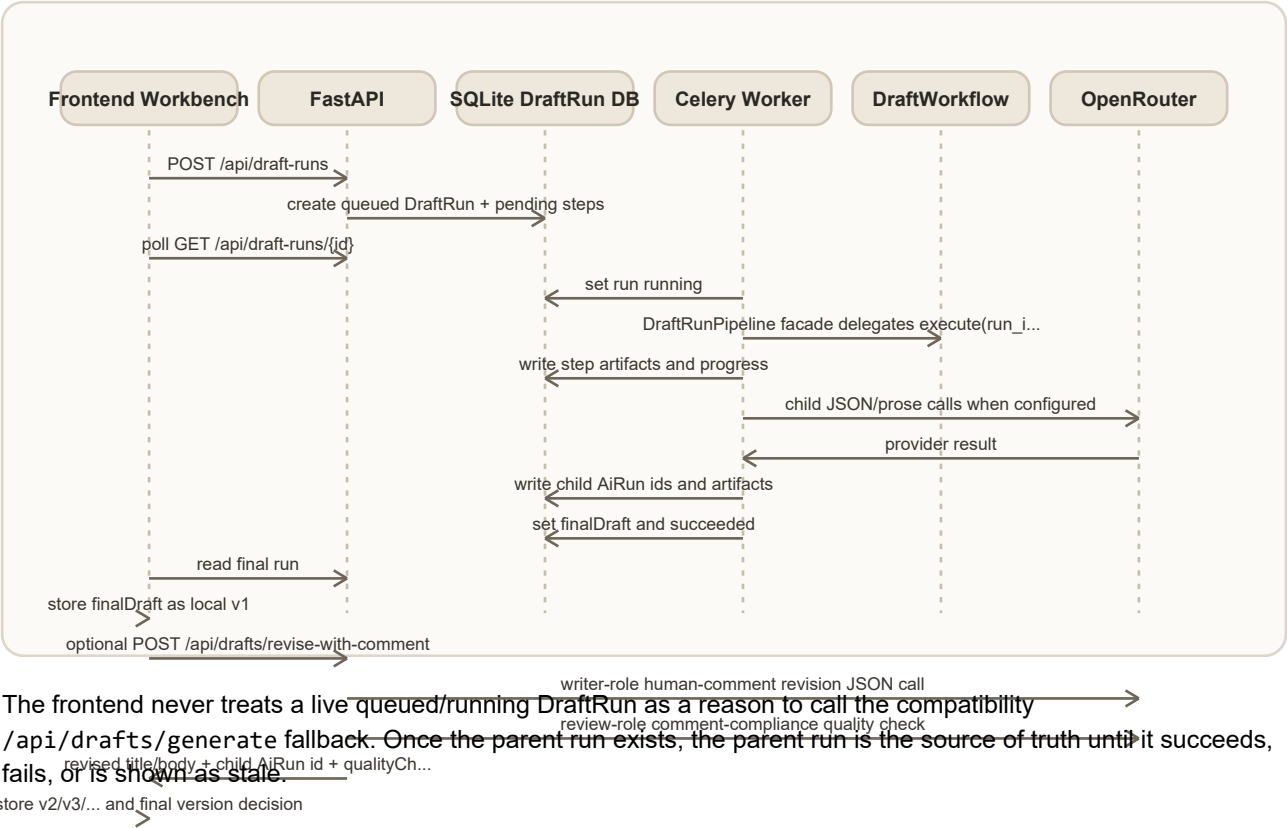
Concept	Meaning	Stored where
RevisionLoop	Bounded editorial optimization loop: validator repair goals plus editorial goals, old-vs-new dimension scoring, accepted/rejected revisions, and rejected moves.	validation.rankingRevision.revisionLoop
FinalQualityGate	Last machine acceptance layer for the delivered final draft as public prose; combines deterministic checks with an independent final-gate model review and may run bounded writer repair cycles if actionable gate findings require it. It separates real missing attribution from diagnostic attribution handoff noise.	validation.rankingRevision.finalQualityGate
DraftRunBudget	Effective research/execution caps derived from Fabula.researchDepth and backend execution mode.	context.draftRunBudget, downstream budget metadata
PayloadBudget	Provider-input boundary for LLM calls: operation profile, execution mode, prompt/token estimates, sent/trimmed counts, suppressed fields, semantic inputs, quality risk, and over-budget incidents.	child AiRun.requestPayload.payloadBudget, attempts, operationEnvelope.payloadStats
ValidationRuntimeBudget	Runtime boundary for the validation/revision loop: wall-clock, LLM-call, revision-cycle, pairwise-round, final-repair, non-improvement, heartbeat, slow-but-healthy, and canonical stop-reason accounting.	validation.progress.runtimeBudget, validation.rankingRevision.runtimeBudget, validation.rankingRevision.revisionLoop.runtimeBudget
QualityFidelityReport	Final per-run distinction between technical completion, provider retry/backup/fallback recovery, evidence fidelity, validation/final-gate issue lifecycle, editorial publishability, and overall clean/degraded/attention verdict. A succeeded DraftRun is not trusted quality by itself: open critical issues block clean success, and unresolved final-gate warnings can only be publishable with caution.	validation.rankingRevision.qualityFidelity, complete.qualityFidelity, diagnostics helper output
ProviderReliabilityReport	Cross-run provider reliability summary and remediation map for retry, backup, fallback, timeout, malformed JSON, schema failure, payload/runtime budget, degraded, failed, and open critical patterns.	python scripts/analyze_draft_run_reliability.py --run-id ...
finalDraft	The selected draft returned to the frontend after validation, ranking, revision loop, and final quality gate.	parent DraftRun
DraftVersion	Immutable editor-facing version of the delivered draft. v1 is the machine final draft; later versions come from human-comment revisions or manual edits.	local workspace postDraft.versions[]
HumanCommentRevisionQualityCheck	Diagnostic review of one human-comment revision: comment compliance, source-marker preservation, public-prose health, internal jargon leaks, and base-version regression risks. It never blocks saving the version.	local workspace postDraft.versions[].qualityCheck, child AiRun trace
EditorDecisionSnapshot	Human final-selection record linking the chosen version to machine trace summaries, unresolved risks, comments, and manual edit counts.	local workspace finalText.editorDecisionSnapshot

Concept	Meaning	Stored where
EditorialLearningAuthorNote	Auto-created author-memory note summarizing what the editor appears to have taught the system through final version choice, comments, manual edits, rejected versions, and quality checks. Starts as pendingReview; only accepted notes influence author-position inference.	local workspace authorNotes[], type editorialLearning

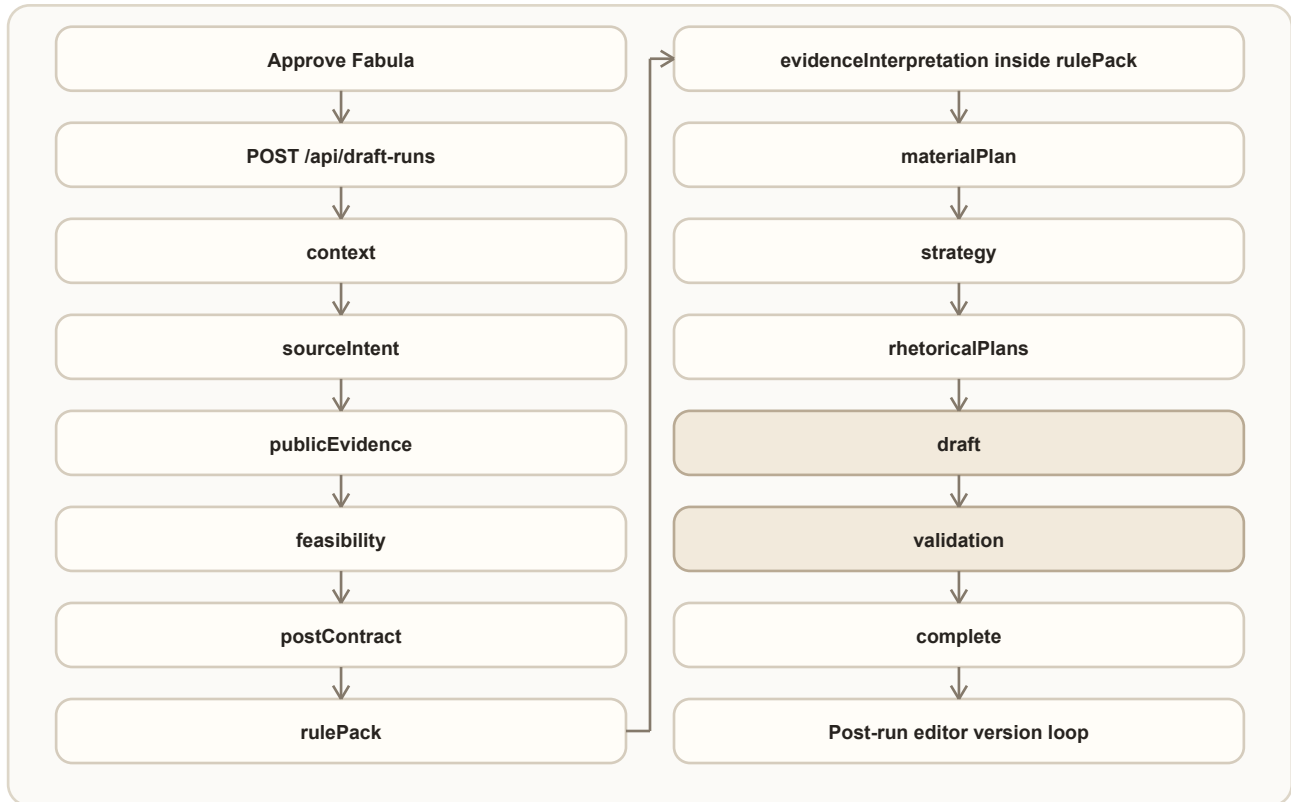
2. Runtime topology

Runtime entrypoints are compatibility-preserving:

- the Celery/API path still enters the drafting runtime through `backend.app.application.draft_run_pipeline.DraftRunPipeline;`
- `DraftRunPipeline` is now a thin facade that keeps the previous constructor and `execute(run_id)` method;
- sequencing is delegated to `backend.app.drafting.application.workflow.DraftWorkflow` through an ordered `DraftStepRegistry`;
- context/artifact, source/evidence, evidence-contract, planning, generation, validation, revision, final-quality, and HITL services now have canonical owners under `backend.app.drafting.application`;
- old `backend.app.application.*` imports for migrated runtime modules are thin compatibility shims only;
- prompt text, provider calls, artifact payloads, progress semantics, child `AiRun` persistence, and blocked/failure behavior are unchanged.



3. Current step order



The backend enum is:

context -> sourceIntent -> publicEvidence -> feasibility -> postContract -> rulePack -> materialPlan -> strategy -> rhetoricalPlans -> draft -> validation -> complete

The ordered runtime registry uses the same sequence. It is a package-boundary refactor, not a semantic step change.

4. Step-by-step technical flow

The role system is not a separate chat room. Roles interact through persisted DraftRun artifacts. Each LLM call chooses a role model through DraftModelRoleResolver, receives the artifact set and the role-specific ContextPack, writes a child AiRun, and stores its result back into the parent step artifact. The next role reads that artifact instead of relying on hidden conversation state.

Role-aware execution map:

Step	Active role	Model selection	Context passed to the role	Output consumed by
context	none	no model	frontend snapshot only	source intent, ledger, feasibility, DraftRun budget
sourceIntent	research	DRAFT_RESEARCH_MODEL, then default	brief sources, context summary, initial ledger, DraftRun budget	public evidence
publicEvidence search	web search	OPENROUTER_WEB_SEARCH_MODEL	budget-capped research tasks and built search query	evidence synthesis
publicEvidence synthesis	research	DRAFT_RESEARCH_MODEL, then default	accepted public evidence, initial ledger, context	enriched ledger, dossier

Step	Active role	Model selection	Context passed to the role	Output consumed by
feasibility	none	no model	enriched ledger and warnings	contract or blocked completion
postContract	none	no model	brief, ledger, feasibility, size settings	rules, planning, validation
rulePack	none	no model	contract, ledger, publisher/topic/fabula rules	evidence interpretation, validation, revision
evidenceInterpretation inside rulePack	strategy	DRAFT_STRATEGY_MODEL, then default, repair, backup	compact enriched ledger, evidence synthesis, accepted public evidence, contract, relevant rule subset, strategy ContextPack, timeout envelope	material plan, dossier, context packs
materialPlan	strategy	DRAFT_STRATEGY_MODEL, then default, repair, backup	strategy ContextPack, usable evidence candidates, evidence interpretation, contract, registry	draft strategy
strategy	strategy	DRAFT_STRATEGY_MODEL, then default	material plan, rules, contract, strategy ContextPack	rhetorical plans
rhetoricalPlans	strategy	DRAFT_STRATEGY_MODEL, then default, repair, backup	strategy, material plan, claim/rule ids	writer candidates
draft	writer	DRAFT_WRITER_MODEL, then default, repair, backup, domain-safe fallback	writer ContextPack, one rhetorical plan, material evidence, contract, writer generation params	validation and ranking
validation lint	none	no model	candidates, contract, registry, ledger	LLM validation and ranking
validation LLM review	review	DRAFT_REVIEW_MODEL, then default, repair, backup	review ContextPack, candidates, deterministic findings	pairwise ranking
validation editorial critique	critic	DRAFT_CRITIC_MODEL, then default, repair, backup	critic ContextPack, candidates, evidence interpretation, validation findings	trace, dossier, future critique-aware ranking
validation alternative angle	anotherAngle then writer	DRAFT_ANOTHER_ANGLE_MODEL for route, DRAFT_WRITER_MODEL for challenger prose, repair, backup, no fake challenger fallback	initial validation, editorial critique, another-angle ContextPack, writer ContextPack, another-angle/writer generation params	final validation and ranking
validation ranking	review	DRAFT_REVIEW_MODEL, then default, repair, backup	merged candidates, old scorecard, deterministic and LLM findings	revision loop
validation revision	writer	DRAFT_WRITER_MODEL, then default, repair, backup, failed/not-run if no usable JSON	current best candidate, repair goals, anti-regression constraints, writer ContextPack, revision generation params	regression guard and final draft
validation final quality gate	finalGate, then optional writer	DRAFT_FINAL_GATE_MODEL, or independent critic/review/default fallback; DRAFT_WRITER_MODEL for bounded final repair cycles	delivered final candidate, FinalQualityContract, validation reports, critique, evidence interpretation, ledger, contract, rule registry, material plan	accepted public draft or rejected repair trace
complete	none	no model	final DraftRun state	frontend

Step	Active role	Model selection	Context passed to the role	Output consumed by
post-run human comment revision	writer	DRAFT_WRITER_MODEL, then default, repair, backup, failed if no usable JSON	selected DraftVersion, editor comment, compact machine trace context from final quality gate, revision loop, alternative angle, validation, contract/material evidence	new local DraftVersion; not a new DraftRun
post-run human revision quality check	review	DRAFT_REVIEW_MODEL, then default, repair, backup, notRun if no usable JSON	base version, revised version, editor comment, compact machine trace context, source markers and public-prose constraints	DraftVersion.qualityCheck; child AiRun, not a DraftRun step

Practical reading rule: when a step says Role/model handoff: strategy, it does not mean that the strategy model talks directly to the writer model. It means the strategy role writes a structured artifact, then the writer role later receives that artifact plus its own compact context pack.

4.1 context

Purpose: build the normalized local context for the selected work item.

Input:

- approved PostBrief;
- EditorialModel;
- draftContext snapshot from frontend: work item, plan slot, candidate, signal, topic, fabula, publisher rules, author-position evidence, publication-size context.

Processing:

- build_draft_run_context_summary(...) normalizes the frontend snapshot;
- SourceLedgerBuilder builds the initial internal claim ledger.
- DraftRunBudgetResolver combines fabula.researchDepth with DRAFT_RUN_EXECUTION_MODE.

Output:

- contextSummary;
- initial sourceLedger;
- draftRunBudget;
- missingContext and compatibility metadata when links are absent.

Role/model handoff:

- no LLM role is used;
- this step creates the first artifact boundary for later roles;
- downstream research, strategy, writer, and review calls never read the raw frontend workspace directly after this point.

Trace: steps[].stepKey = context.

4.2 sourceIntent

Purpose: convert approved brief sources into an explicit research plan.

Input:

- `PostBrief.sources`;
- context summary;
- initial `SourceLedger`.

Processing:

- `SourceIntentNormalizer` classifies URLs, named sources, human-language requests, proof needs, framing hints, exclusions, and unknown lines;
- `ResearchPlanService` may call `OpenRouter` through the research role;
- deterministic fallback preserves original wording when provider planning fails;
- `source_research_budgeting` caps executable verification tasks and records `budgetTrace.budgetSkipped` instead of silently dropping them.

Output:

- normalized `sourceIntent`;
- `researchPlan`;
- `draftRunBudget` and `budgetTrace`;
- child `AiRun` when `OpenRouter` is used.

Role/model handoff:

- active role: research;
- model resolver chooses `DRAFT_RESEARCH_MODEL` or falls back to `OPENROUTER_DEFAULT_MODEL`;
- the role receives sources, context summary, and the initial ledger;
- it returns a `ResearchPlan`, not proof;
- `publicEvidence` consumes the plan and decides what can actually be read or searched.

Trace: `sourceIntent` step and child `AiRun.requestPayload.draftRunStep` = `sourceIntentResearchPlan`.

4.3 publicEvidence

Purpose: execute available public evidence tasks and enrich the ledger.

Input:

- `sourceIntent` artifact;
- `researchPlan`;
- current context artifact.

Processing:

- exact URL tasks are read by the URL reader;
- public search tasks call OpenRouter web search only when web tools are enabled;
- retrieval tasks, URL reads, and search result counts are capped by DraftRunBudget;
- disabled or unavailable search tasks remain explicit attempts, not proof;
- relevance guard rejects search-result drift;
- EvidenceSynthesis reconciles accepted public evidence into external claims;
- SourceLedgerExternalEvidenceMerger merges accepted external claims into the ledger;
- accepted evidence items and external ledger claims are trimmed by budget before downstream planning;
- operation progress is persisted during URL/search/synthesis work.

Output:

- PublicEvidenceBatch: attempts, accepted evidence items, warnings;
- EvidenceSynthesis: external claims, decisions, warnings;
- enriched SourceLedger;
- budget metadata: used counts, cap hits, skipped tasks, trimmed evidence/claims;
- initial ArticleDossier and all role ContextPacks.

Role/model handoff:

- public search uses the web-search model/config;
- evidence synthesis uses research.
- web search receives concrete URL/search tasks and built search queries;
- research synthesis receives accepted evidence candidates and decides how they should become external ledger claims;
- downstream roles consume the enriched ledger and dossier, not raw search output.

Trace:

- publicEvidence artifact;
- child AiRun.requestPayload.draftRunStep = externalEvidenceSynthesis;
- child web-search AiRun records when search is enabled.

AS IS limitation:

- public evidence retrieval and synthesis do not write prose directly. Accepted evidence is merged into the ledger here, then interpreted inside rulePack before material planning and writing.

4.4 feasibility

Purpose: decide whether it is safe to generate prose.

Input:

- enriched SourceLedger;
- context warnings;

- evidence availability.

Processing:

- deterministic quality gate classifies the run as feasible, feasible with constraints, needs research, needs human decision, or infeasible.

Output:

- FeasibilityReport;
- if blocked: `complete.status = blocked`, `finalDraft = null`, no local fallback.

Role/model handoff:

- no LLM role is used;
- this deterministic gate decides whether later roles are allowed to write prose;
- if blocked, `strategy`, `writer`, and `review` are not called.

Trace: feasibility step.

4.5 postContract

Purpose: lock what the post must preserve.

Input:

- approved brief;
- context summary;
- enriched SourceLedger;
- feasibility result;
- publication-size profile and fabula size intent.

Processing:

- PostContractBuilder resolves title, thesis, audience, value, goal, CTA, platform/date/time, topic/fabula ids, allowed claim ids, qualified claims, forbidden moves, evidence obligations, fabula obligations, risks, and PublicationSizeContract.

Output:

- provider-free PostContract.

Role/model handoff:

- no LLM role is used;
- this step creates the invariant contract that all later roles must preserve;
- strategy may plan within the contract, writer may execute it, and review evaluates candidates against it.

Trace: postContract step.

4.6 rulePack

Purpose: turn the contract and editorial rules into machine-readable rules.

Input:

- enriched context artifact;
- SourceLedger;
- FeasibilityReport;
- PostContract;
- topic/fabula/publisher rules.

Processing:

- rulePack is marked running before provider-backed evidence interpretation, so a long JSON retry path is visible as rulePack, not as stale postContract;

- RuleRegistrySnapshot is compiled first;
- compatibility RulePack is derived from the registry;
- the provider request is compacted before JSON generation: full ruleRegistrySnapshot

stays in the artifact, while the model receives relevant hard/critical/evidence/ style/attribution/topic/fabula rules, accepted public evidence summaries, external ledger claims, evidence synthesis, and the locked contract;

- each provider attempt runs inside DRAFT_EVIDENCE_INTERPRETATION_TIMEOUT_SECONDS

so a stuck strategy-model call can fail, write a child AiRun, and continue to repair/backup/fallback instead of leaving rulePack running forever;

- EvidenceInterpretationService converts accepted internal/external evidence into

editorial implications, tensions, examples, limits, forbidden overclaims, reader value hooks, and rejected evidence uses before material planning;

- EvidenceInterpretationFidelityPolicy classifies the interpretation as

sufficient, partial, weak, or missing. Retry/repair and backup acceptance are recovery signals, not evidence degradation by themselves. Deterministic fallback always lowers evidence fidelity, and missing accepted evidence blocks a trusted editorial verdict.

Output:

- ruleRegistrySnapshot;
- compatibility rule-pack fields;
- evidenceInterpretation;
- evidenceInterpretationFidelity;
- attempts[] for the interpretation provider/repair/backup path;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- rule compilation itself uses no LLM role;
- evidence interpretation uses the strategy role through DRAFT_STRATEGY_MODEL, then default, repair, optional backup, and finally deterministic fallback;
- this step translates editorial obligations into machine-readable rule ids and then

translates evidence into usable editorial meaning;

- later role prompts receive rule ids and validator bindings so they do not work

from anonymous prose constraints only;

- material planning and writing receive interpretation artifacts instead of raw

citation snippets as their main source context.

Trace:

- rulePack step;

- child `AiRun.requestPayload.draftRunStep` = `evidenceInterpretation` when provider interpretation runs;

- `artifactPayload.progress.operations[ ]` contains `evidenceInterpretation` primary/repair/backup attempts while the step is running;

- child `AiRun.requestPayload.inputStats` records original/compact rule counts, claim counts, evidence counts, prompt char estimate, selected model, and timeout seconds;

- attempt records may include `status=timeout` plus operation timing notes such as prompt build, provider request, JSON/shape validation, and `AiRun` persistence;

- `rulePack.evidenceInterpretation` readable section in `/ai-runs?runId=...`

4.7 materialPlan

Purpose: decide what material is usable before strategy and writing.

Input:

- context summary;
- RulePack and RuleRegistry;
- EvidenceInterpretation;
- enriched SourceLedger;
- PostContract;
- strategy ContextPack.

Processing:

- material planner receives `usableEvidenceCandidates`;
- the candidate list is capped by `DraftRunBudget.maxUsableEvidenceCandidates`;
- material planner receives `evidenceInterpretation` and should prefer implications, usable examples, limits, and forbidden overclaims over raw public snippets;
- it must either choose usable evidence or explain rejection reasons;
- accountability guard rejects empty/unexplained evidence selection;
- retry sequence: primary -> repair -> optional backup model -> emergency deterministic fallback.

Output:

- MaterialPlan;
- availableEvidence;

- rejectedEvidence;
- claimsRequiringAttribution;
- qualifiedClaims;
- attempts[];
- accountability metadata;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- active role: strategy;
- model resolver chooses DRAFT_STRATEGY_MODEL, then default, then repair/backup

where applicable;

- the role receives the strategy ContextPack, usable evidence candidates,

ledger claim ids, contract, and registry;

- it must return selected or rejected evidence with reasons;
- strategy and rhetoricalPlans consume the material plan instead of scanning the whole ledger again.

Trace:

- materialPlan step;
- child AiRun.requestPayload.draftRunStep = materialPlan.

4.8 strategy

Purpose: define the main strategy for the post.

Input:

- context summary;
- RulePack;
- MaterialPlan;
- strategy ContextPack.

Processing:

- OpenRouter JSON call or deterministic fallback builds the draft strategy.

Output:

- DraftStrategy: thesis angle, opening move, argument sequence, fabula usage, CTA plan, forbidden moves, tone notes;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- active role: strategy;
- model resolver chooses DRAFT_STRATEGY_MODEL or default;

- the role receives MaterialPlan plus the strategy ContextPack;
- it creates the main route of argument, but cannot change the PostContract;
- rhetoricalPlans expands this strategy into several executable routes.

Trace: strategy step and child AiRun.requestPayload.draftRunStep = strategy.

4.9 rhetoricalPlans

Purpose: build several routes for executing the same contract.

Input:

- context summary;
- RulePack;
- MaterialPlan;
- DraftStrategy.

Processing:

- one JSON call returns 2-3 rhetorical plans;
- retry sequence: primary -> repair -> optional backup -> deterministic fallback.

Output:

- RhetoricalPlanSet;
- plan moves, claim ids to use/avoid, required rule ids, CTA route, risks;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- active role: strategy;
- model resolver chooses DRAFT_STRATEGY_MODEL, then repair/backup if JSON shape

is invalid;

- the role receives contract, material plan, draft strategy, claim ids, rule ids,

and the strategy ContextPack;

- it returns alternative rhetorical plans that the writer must execute;
- the writer must not invent a new route outside these plan artifacts.

Trace:

- rhetoricalPlans step;
- child AiRun.requestPayload.draftRunStep = rhetoricalPlans.

4.10 draft

Purpose: generate candidate drafts from rhetorical plans.

Input:

- request brief;

- context summary;
- RulePack;
- MaterialPlan;
- DraftStrategy;
- RhetoricalPlanSet;
- EvidenceInterpretation;
- writer ContextPack.

Processing:

- one candidate is generated per rhetorical plan;
- the executed candidate count is capped by `DraftRunBudget.maxDraftCandidates` before provider calls are attempted;
- each provider failure falls back per candidate, not for the whole step;
- candidate publishability guard marks candidates eligible, penalized, or excluded;
- deterministic v1 selection creates an initial scorecard.

Output:

- candidates[];
- selection;
- scorecard;
- child AiRun ids;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- active role: writer;
- model resolver chooses `DRAFT_WRITER_MODEL` or default;
- each candidate call receives one rhetorical plan, writer ContextPack, material evidence, evidence interpretation, allowed claim ids, forbidden moves, and size contract;
- each JSON candidate attempt follows the universal retry sequence: primary writer model, repair prompt, optional backup model, then domain-safe deterministic fallback only if provider attempts are exhausted;
- primary writer attempts use `DRAFT_WRITER_TEMPERATURE` and `DRAFT_WRITER_TOP_P`;
- repair and backup attempts use `DRAFT_JSON_REPAIR_TEMPERATURE`, not the creative writer temperature;
- writer prompts forbid leaking internal artifact names such as `SourceLedger`, `publicEvidence`, `validators`, `RuleRegistry`, or `PostContract` as unexplained public dev-jargon;
- candidates write prose, but selection still remains downstream responsibility;
- review receives all candidates plus validation context.

Trace:

- draft step;

- child `aiRun.requestPayload.draftRunStep` = `draftCandidate`.

AS IS behavior:

- writer prompts now receive interpreted implications, examples, limits, and forbidden overclaims, so public evidence should shape the argument instead of being pasted as a decorative citation block;
- each provider attempt is visible through child `aiRun` audit with model role, selected model, attempt label, repair/backup marker, generation params, and safe validation error.

4.11 validation

Purpose: validate candidates, rank them, revise the selected candidate, and choose the final draft.

Input:

- draft artifact;
- context artifact;
- `RulePack` and `RuleRegistry`;
- `MaterialPlan`;
- review/writer context packs through child services.

Processing:

1. deterministic lint checks all candidates;
2. LLM validation checks all candidates and separates actionable findings from observations;
3. editorial critique challenges all candidates for idea strength, tension, author stance, source integration, generic prose, and reader value;
4. alternative-angle tournament may add one critic-driven challenger candidate; both route JSON and challenger-prose JSON use primary/repair/backup attempts, and no fake deterministic challenger is invented when all attempts fail;
5. final validation report is extended when a challenger enters the pool: initial candidate reports are reused, provider-heavy validation runs only for the new challenger, and the reports are merged into one final candidate-pool report;
6. pairwise ranking chooses the best candidate from the merged candidate pool;
7. deep revision loop builds validator and editorial goals, revises, re-runs deterministic validation, compares previous best vs revised across editorial dimensions, and accepts only targeted improvement without regression. Accepted cycles record concrete reasons such as resolved goals, clear pairwise win, and no deterministic regression;
8. final quality gate builds a `FinalQualityContract` from Fabula research depth, publication kind, post contract, rule registry, and material plan. It checks the delivered candidate with deterministic public-prose rules and an independent `finalGate` model review. If it leaks internal pipeline names, reads like a source dump, loses author stance relative to the contract, or misses reader value, the gate creates bounded final repair instructions for the writer. The number of final repair cycles is controlled by `DRAFT_FINAL_REPAIR_MAX_ITERATIONS` (default 2). A repaired draft is accepted only when deterministic regression and gate findings improve. Attribution warnings are actionable only when they resolve to source-backed claim ids with expected markers that are missing from the final body. Free-text attribution requirements that cannot be resolved become diagnostic handoff noise; if the independent final-gate review passes source integration, that diagnostic noise does not launch final repair;

9. validation and final-gate warnings/critical findings are folded into

`qualityFidelity.issueLifecycle`. Each item is classified as `resolved`, `suppressed`, `acceptedRisk`, or `open`. Accepted final repair resolves the gate issue it repaired. Explicit suppression or accepted risk remains visible in trace. Open critical findings block trusted editorial quality, and an unresolved final-gate warning prevents `cleanSuccess` even when `DraftRun.status` is `succeeded`;

10. provider-heavy validation sub-operations persist progress as they run; if a late

operation fails, the operation is marked `failed`, safe error details are saved, and the previous best candidate is kept when one exists.

Output:

- deterministic validation report;
- `llmValidationReport`;
- `editorialCritiqueReport`;
- `alternativeAngleTournament`;
- `rankingRevision`;
- `revisionLoop` with editorial goals, dimension scores, rejected moves, and stop

reason;

- `finalQualityGate` with final quality contract, independent review attempts,

final-draft status, public-prose status, source-dump risk, actionable attribution findings, diagnostic attribution noise, attribution review closure, final repair goals, repair cycles, repair result, and gate-level final decision;

- `qualityFidelity.issueLifecycle` with lifecycle status and reason for validation

and final-gate warning/critical issues, including synthetic final-gate items when the gate status is `warning` or `critical`;

- progress with nested validation operations, child `AiRun` ids, failed-operation

status, and safe error details when a provider-heavy sub-operation fails;

- final draft candidate decision;
- updated `ArticleDossier` and `ContextPacks`.

Role/model handoff:

- review for LLM validation and pairwise ranking;
- critic for report-only editorial critique;
- `anotherAngle` for challenger route generation;
- `finalGate` for independent final public-prose acceptance;
- writer for directed revision and final public-prose repair.
- deterministic lint runs first without a model;

review receives candidates, review `ContextPack`, deterministic findings, rules, ledger, and contract, then returns report-only findings and pairwise ranking;

- critic receives candidates, critic `ContextPack`, evidence interpretation, and validation context, then returns report-only critique findings and observations;

`anotherAngle` receives the initial validation, editorial critique, dossier cards, and another-angle `ContextPack`, then returns one different contract-safe route;

- writer executes the alternative route into one challenger candidate;

- finalGate receives the delivered final candidate, FinalQualityContract, deterministic gate findings, validation context, evidence interpretation, ledger, rule registry, material plan, and attribution requirement resolution, then returns independent findings, observations, and final repair goals;
- writer may execute bounded final public-prose repair cycles after the revision

loop if finalQualityGate returns warning or critical;

- writer receives the current best candidate plus validator repair goals, editorial goals, rejected moves, dossier/context pack, and constraints, then returns revised prose;
- directed revision attempts use DRAFT_REVISION_TEMPERATURE and DRAFT_REVISION_TOP_P; malformed-JSON repair and backup attempts use the JSON repair temperature;
- regression guard checks deterministic health, and review compares old-vs-new by idea strength, tension, reader value, author stance, source integration, structure, and validator health;
- acceptance policy decides whether the revised prose replaces the current best;
- complete receives only the final decision, not the whole model conversation.

Trace:

- validation step;
- child AiRun.requestPayload.draftRunStep = llmValidation;
- child AiRun.requestPayload.draftRunStep = editorialCritique;
- child AiRun.requestPayload.draftRunStep = alternativeAngleRoute;
- child AiRun.requestPayload.draftRunStep = alternativeAngleCandidate;
- child AiRun.requestPayload.draftRunStep = pairwiseRanking;
- child AiRun.requestPayload.draftRunStep = directedRevision.

4.12 complete

Purpose: mark the run result.

Output cases:

- normal success: parent DraftRun has finalDraft;
- quality-blocked success: parent DraftRun has finalDraft = null and complete.status = blocked;
- failure: parent DraftRun status is failed with a safe error.

Role/model handoff:

- no LLM role is used;
- this step only records the final orchestration outcome for the frontend and

trace workbench.

4.13 Post-run editor version loop

Purpose: let the human editor keep improving or manually editing the delivered machine draft without mutating older versions.

Inputs:

- parent `DraftRun.finalDraft`, stored locally as `DraftVersion v1` with `source=machineFinal`;
- active draft version chosen by the editor;
- optional editor comment;
- optional compact machine trace context from the parent `DraftRun`: final quality gate summary, revision loop summary, alternative-angle outcome, validation summary, unresolved risks, post contract, and material evidence.

Behavior:

- the editor can switch between saved versions;
- manual text edits are local until the user saves them as a new `source>manualEdit` version;
- one click on `Улучшить по комментарию` calls `POST /api/drafts/revise-with-comment`;
- that endpoint is not a new `DraftRun`. It is a post-run writer-role JSON call with child `AiRun` audit and existing JSON retry discipline;
- after a successful writer revision, the endpoint runs a lightweight review role quality check. It evaluates whether the revised version followed the editor comment, preserved visible source markers from the base version, avoided internal pipeline jargon, and did not regress as public prose;
- quality check failure or provider unavailability does not cancel the created version. The version receives `qualityCheck.status = notRun` with attempt metadata;
- provider failure does not create a fake version;
- the editor can mark any saved version as final, including `v1` after later versions exist;
- final version selection creates or updates one deterministic editorialLearning note in Author Memory. The note is visible immediately with status `pendingReview`, but it does not influence author-position inference until the editor accepts it.

Output:

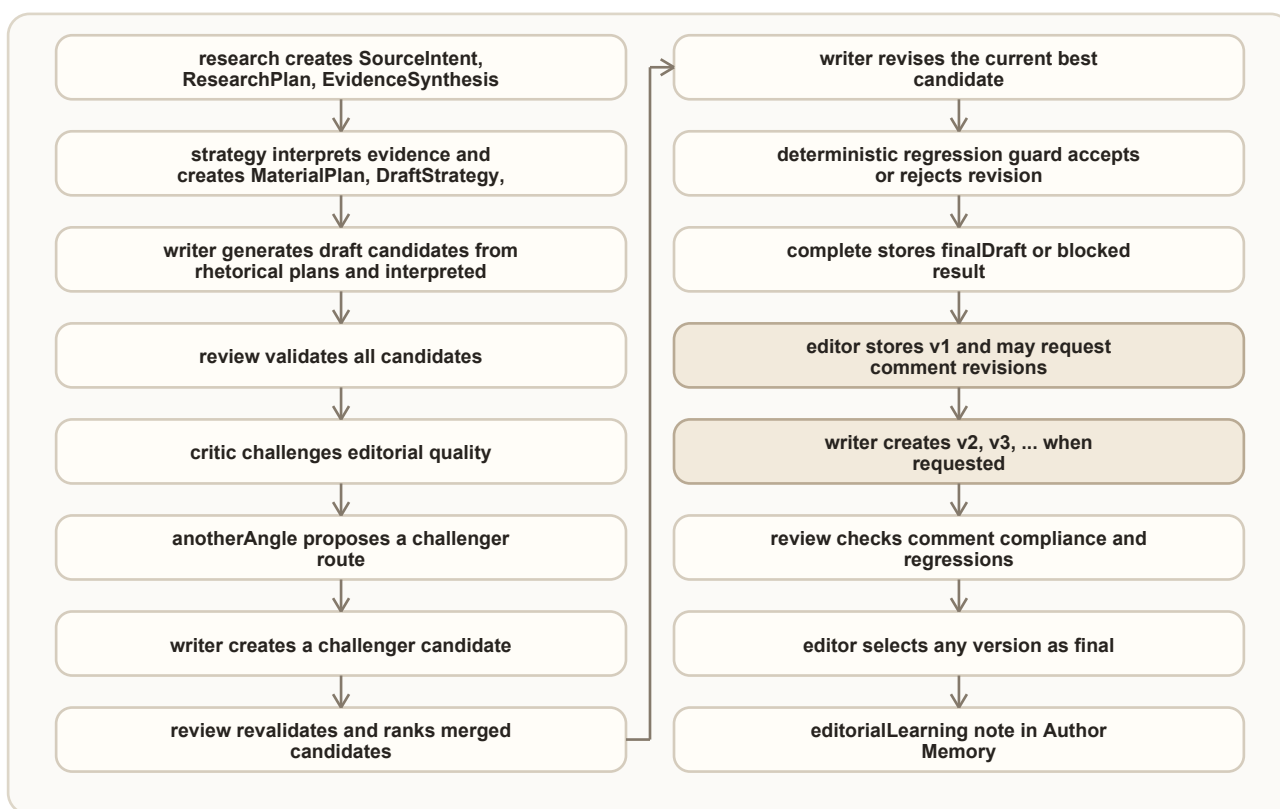
- local `PostDraft.versions[]`, `activeVersionId`, and `finalVersionId`;
- compatibility mirror fields `PostDraft.title/body/version` always reflect the active version;
- `FinalText.draftVersionId`, `FinalText.versionNumber`, and `FinalText.editorDecisionSnapshot`;
- `DraftVersion.qualityCheck` for human-comment revisions when the review ran or explicitly could not run;
- one `AuthorNote.type = editorialLearning` with selected/rejected version metadata, comment summaries, manual edit count, quality-check summaries, unresolved risks, suggested takeaway, and status `pendingReview | accepted | rejected`.

Role/model handoff:

- post-run human-comment revision uses the writer role;
- it receives a compact machine context, not raw DraftRun JSON;
- it must preserve source markers, allowed claims, forbidden moves, and final quality constraints.
- post-run human revision quality check uses the review role;
- it is diagnostic: it can mark the version as passed, warning, critical, or not-run, but it cannot block saving or final approval.

5. Role model usage and handoff AS IS

The current role interaction is artifact-mediated. A role does not continue the previous model's hidden conversation. It receives explicit artifacts and a role-specific context pack, then writes a new artifact for the next role.

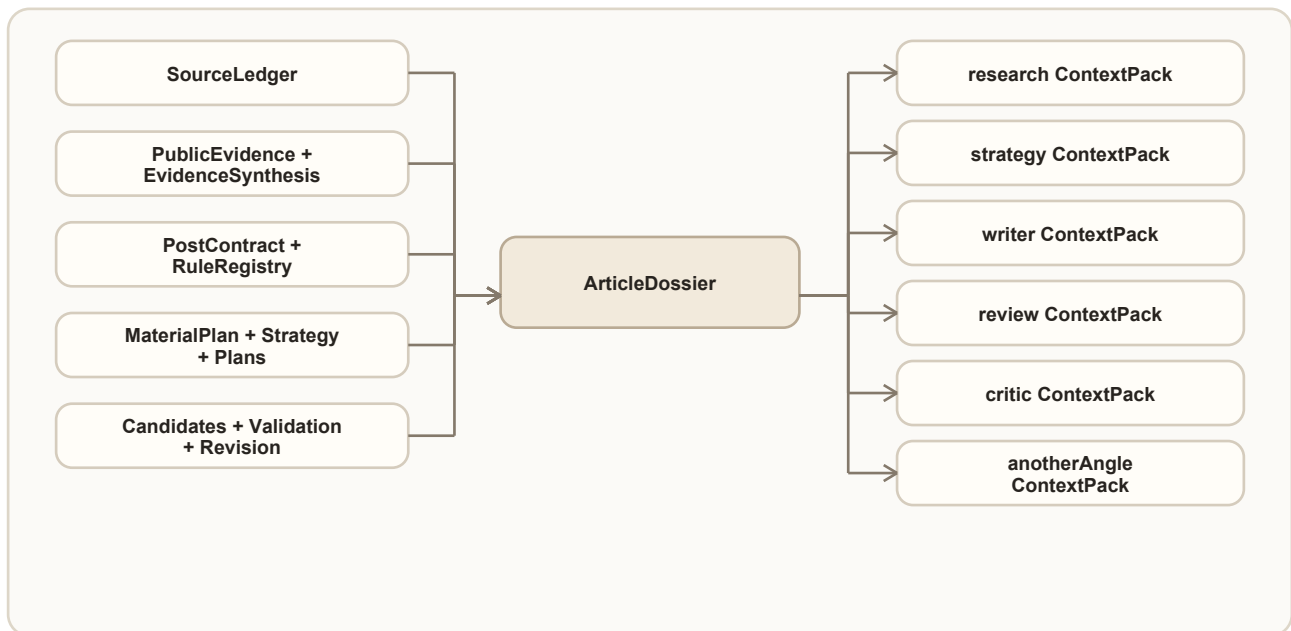


Role summary:

Role	Used now	Receives	Must not do
research	source research planning and evidence synthesis	source intent, public evidence, context	invent facts or strengthen weak evidence
strategy	evidence interpretation, material plan, draft strategy, rhetorical plans	strategy context pack, rules, contract, material, evidence synthesis	change post thesis or bypass contract
writer	draft candidates and directed revisions	writer context pack, evidence interpretation, material plan, rhetorical plan, repair goals	create new claims or violate forbidden moves
review	LLM validation and pairwise ranking	review context pack, candidates, rules, ledger, validation reports	rewrite prose directly
critic	report-only editorial critique	critic context pack, evidence interpretation, candidates, validation context	rewrite prose or change final selection

Role	Used now	Receives	Must not do
anotherAngle	one critic-driven challenger route	another-angle context pack, initial validation, editorial critique, rejected/weak moves	act as technical backup or invent new evidence
writer post-run	human-comment revision of one saved version	active draft version, editor comment, compact machine trace context	mutate prior versions or fabricate a successful revision after provider failure
review post-run	comment-compliance quality check for human-comment revisions	base version, revised version, editor comment, compact machine trace context	block saving a successful revision or turn warnings into learning signals automatically

6. Context flow



Important AS IS rules:

- raw DraftRun trace is not the normal prompt context;
- ArticleDossier is rebuilt from selected artifacts;
- ContextPack selects up to 12 cards for one role;
- child AiRun.requestPayload stores the role model and context pack it received;
- context packs are local orchestration artifacts, not long-term memory.

7. Artifact catalog

Artifact	Created by	Read by	Debug location
SourceLedger	context/publicEvidence	feasibility, contract, rules, planning, validation	context.sourceLedger, publicEvidence.enrichedSourceLedger
SourceIntent	sourceIntent	publicEvidence	sourceIntent step
ResearchPlan	sourceIntent	publicEvidence	sourceIntent.researchPlan
PublicEvidenceBatch	publicEvidence	evidence synthesis, trace	publicEvidence.items, publicEvidence.attempts
EvidenceSynthesis	publicEvidence	ledger merger, material planning, trace	publicEvidence.evidenceSynthesis

Artifact	Created by	Read by	Debug location
DraftRunBudget	context	sourceIntent, publicEvidence, material plan, draft, validation	context.draftRunBudget, step budget metadata
EvidenceInterpretation	rulePack	material plan, writer, dossier, trace	rulePack.evidenceInterpretation
PostContract	postContract	rules, planning, validation, revision	postContract step
RuleRegistrySnapshot	rulePack	material plan, validation, revision	rulePack.ruleRegistrySnapshot
MaterialPlan	materialPlan	strategy, writer, validation, ranking	materialPlan.materialPlan
DraftStrategy	strategy	rhetorical plans, writer	strategy.draftStrategy
RhetoricalPlanSet	rhetoricalPlans	draft candidates	rhetoricalPlans.rhetoricalPlanSet
DraftCandidate	draft	validation, ranking, revision	draft.candidates[]
ValidationReport	validation	ranking/revision, trace	validation.candidateReports
LlmValidationReport	validation	ranking/revision, trace	validation.llmValidationReport
EditorialCritiqueReport	validation	alternative-angle tournament, trace, dossier	validation.editorialCritiqueReport
AlternativeAngleTournament	validation	final validation/ranking, trace	validation.alternativeAngleTournament
RankingRevision	validation	final draft selection, trace	validation.rankingRevision
FinalQualityGate	validation	final draft selection, trace	validation.rankingRevision.finalQualityGate
QualityFidelityReport	workflow completion	diagnostics, editor-facing risk summary, future reliability analytics	validation.rankingRevision.qualityFidelity, complete.qualityFidelity
ArticleDossier	article memory service	context pack builder, trace	selected step artifacts
ContextPack	article memory service	child LLM services	step artifacts and child AiRun.requestPayload
DraftVersion	frontend/editor actions	editor version list, final selection	local workspace postDraft.versions[]
HumanCommentRevisionQualityCheck	post-run revise-with-comment endpoint	editor version list, final selection, future learning slice	local workspace postDraft.versions[].qualityCheck, child AiRun.requestPayload.draftRunStep = humanCommentRevisionQualityCheck
EditorDecisionSnapshot	final text approval	editorial-learning note builder, audit/debug	local workspace finalText.editorDecisionSnapshot
EditorialLearningAuthorNote	final text approval	author memory feed, optional author-position inference after acceptance	local workspace authorNotes[], type editorialLearning

8. Retry, fallback, and blocked behavior

Area	Behavior
live DraftRun	frontend keeps polling; no compatibility fallback while run is alive
stale DraftRun	diagnostic state after no timestamp progress; not automatic fallback
slow-but-healthy validation	validation has runtimeBudget.currentOperationId and heartbeat/current-operation age inside runtimeBudget.limits.staleAfterSeconds; do not classify as stuck

Area	Behavior
over-budget validation operation	stale diagnostics report the current validation operation and runtime stale budget instead of the old generic five-minute message
public search disabled	attempt is notConfigured; it is not proof
malformed JSON planning	repair retry, optional backup model, then deterministic fallback
malformed, empty, or timed-out evidence interpretation	failed child AiRun, failed nested operation, repair retry, optional backup model, then deterministic interpretation fallback
editorial critique provider missing	editorialCritiqueReport.status=not-run; no fake critique
malformed editorial critique JSON	repair retry, optional backup model, then not-run for that candidate
alternative-angle provider missing	alternativeAngleTournament.status=not-run; original candidates continue
malformed alternative-angle route JSON	repair retry, optional backup model, then tournament failed; no deterministic fake angle
alternative-angle candidate provider failure	repair retry, optional backup model, then tournament failed; original candidates continue
validation sub-operation failure	mark nested operation failed, preserve partial artifact, keep previous best if available
material plan ignores evidence	accountability retry, optional backup model, then emergency fallback
unresolved attribution requirement	record diagnostic metadata; not a repair goal unless it resolves to expected source markers that are missing from the final body
human-comment revision provider failure	no new version is created; UI keeps the current version list and shows a safe error
human revision quality-check failure	created version is kept; qualityCheck.status=notRun records attempts and safe reason
candidate provider failure	fallback only for that candidate direction
fallback candidate selection	fallback is diagnostic unless publishability guard allows it
feasibility blocked	run succeeds with finalDraft = null; no local fallback
all invalid candidates	run succeeds as quality-blocked; no local fallback
revision regression	keep previous best candidate

Slice 2.17.4.6.0.3.2 adds shared operationEnvelope governance for representative provider-heavy operations. Migrated operations expose operationId, operation kind, owner, status, attempts, child AiRun ids, input/payload stats, retry policy, timeout profile where applicable, incident metadata, safe error, and result payload. Fallback, not-run, failed, timeout, cancelled, and stale outcomes require incident metadata. Backup success keeps the accepted payload but records backupAccepted. Representative migrated operations are evidenceInterpretation, editorialCritique, directedRevision, humanCommentRevision, and humanCommentRevisionQualityCheck; other current provider-heavy operations remain in CURRENT_LLM_OPERATION_INVENTORY until their owning migration slices.

Slice 2.17.4.6.0.3.3 adds DraftRun provider-input payload budget governance. Full artifacts stay in parent DraftRun storage, but enforced provider calls receive a curated compact payload immediately before prompt messages are built. The enforced representative operations are evidenceInterpretation, editorialCritique, directedRevision, humanCommentRevision, and humanCommentRevisionQualityCheck. Each records payloadBudget.profileId, execution mode, limits, sent/trimmed counts, suppressed fields, semantic input contract, qualityRisk, promptCharEstimate, and approxTokenEstimate in child AiRun.requestPayload, attempts, and operationEnvelope.payloadStats. If compacted input still exceeds the profile cap, trace records contextOverBudget; hard-cap breaches record payloadTooLarge. Remaining provider-heavy operations are explicit payload-budget debt entries in CURRENT_LLM_OPERATION_INVENTORY.

Slice 2.17.4.6.0.3.4 adds ValidationRuntimeBudget governance for the validation stage. The runtime guard records runtimeBudget.profileId, execution mode, limits, used counters, startedAt, lastHeartbeatAt, currentOperationId, currentOperationStartedAt, slowButHealthy, stopReason, exhausted, and incidents in the existing validation progress artifact. It uses canonical stop reasons: acceptedQuality,

humanReviewRequired, budgetExhausted, maxIterations, noImprovement, and providerIncident; older internal reasons are kept as detailStopReason where useful. No endpoint contract or SQLite schema is changed.

Slice 2.17.4.6.1.0 adds QualityFidelityReport as a trace-only DraftRun artifact. It does not change step order, prompts, provider selection, API, or SQLite schema. The report separates technicalStatus, providerRecoveryStatus, editorialStatus, and overallVerdict. Successful primary repair/retry on the same model is normal recovery, not a quality failure. Backup success is accepted but diagnostic. Domain-safe deterministic fallback lowers fidelity. Step-level quality problems are tracked separately: evidence coverage, unresolved critical/warning findings, final-gate warning/critical status, rejected final repair, and size/over-budget risks.

Slice 2.17.4.6.1.3 adds ProviderReliabilityReport as a developer diagnostics artifact over multiple stored DraftRuns. It reads structured qualityFidelity, operation envelopes, payload/runtime budgets, and child AiRun records; it does not parse prose diagnostics and does not change runtime behavior. One run is reported as insufficientData for systemic conclusions. Every non-clean signal must become a remediation decision such as noActionExpected, watchWithMoreRuns, fixBacklogSlice, or fixBeforeTrustingQuality.

The report also includes signalCoverage. This is the audit layer for the analytics itself: it records raw child AiRun records, operation-envelope incidents, retry/backup/fallback signals, payload/runtime budget incidents, and ignored stats-only budget payloads with reasons. fixBacklogSlice and fixBeforeTrustingQuality remediation decisions must point to concrete roadmap slices.

9. How to read a run trace

Open `/ai-runs?runId=<DraftRun ID>` and inspect in this order:

1. context: confirm source signal, candidate, topic, fabula, rules are present.
2. DraftRun budget: confirm research depth, execution mode, effective caps, used counts, cap hits, skipped tasks, and trimmed evidence/claims.
3. sourceIntent: confirm research requests were normalized correctly.
4. publicEvidence: confirm URL/search attempts, accepted evidence, rejected drift, evidence synthesis, and enriched ledger.
5. feasibility: confirm run was allowed or blocked for quality reasons.
6. postContract: confirm thesis, CTA, size, allowed claims, and forbidden moves.
7. rulePack: confirm rule registry exists and size/evidence rules are present.
8. evidenceInterpretation: inside rulePack, confirm implications, examples, limits, forbidden overclaims, rejected evidence uses, attempts, operationEnvelope, timeout profile, input stats, payloadBudget, and incident taxonomy.
9. materialPlan: confirm selected evidence and explicit rejection reasons.
10. strategy: confirm the strategy does not change the contract.
11. rhetoricalPlans: confirm the plans are different routes, not new topics.
12. draft: compare candidates, source/fallback status, publishability, scorecard.
13. validation: inspect deterministic warnings, LLM findings, observations.
14. editorialCritiqueReport: inspect idea strength, tension, author stance, source integration, generic-prose risks, recommended editorial moves, candidate-level operationEnvelope, malformed JSON/schema incidents, and not-run incidents.
15. alternativeAngleTournament: inspect challenger route, candidate, model attempts,

and why it entered or failed to enter final ranking.

16. `rankingRevision`: inspect pairwise winner, revision cycles, accepted/rejected moves.

Directed revision sub-results should include `operationEnvelope` for accepted, failed, or not-run writer revisions. `revisionLoop.stopReason` is canonical; inspect `detailStopReason` for legacy detail such as editorially-improved, no-fresh-angle, or provider failure detail. Inspect `runtimeBudget` to see max wall-clock, LLM calls, revision cycles, pairwise rounds, final repair cycles, non-improvement count, heartbeat, and slow-but-healthy status.

17. `finalQualityGate`: inspect the final quality contract, independent review

attempts/model, final public-prose status, `attributionReview`, `actionableAttributionFindings`, `diagnosticAttributionNoise`, repair goals, accepted or rejected repair cycles, and final draft source.

18. `qualityFidelity`: inspect `technicalStatus`, `providerRecoveryStatus`,

`evidenceFidelity`, `issuelifecycle`, `editorialStatus`, and `overallVerdict` before equating `DraftRun.status=succeeded` with publication-ready output. 18a. Cross-run reliability: when comparing multiple runs, use `python scripts/analyze_draft_run_reliability.py --run-id <id> --run-id <id>` and inspect `operation/model` counters, `signalCoverage`, plus the remediation ledger. Do not infer a systemic provider/model problem from one successful retry.

19. `validation.progress`: inspect nested operations; failed late operations should show safe errors and the final previous-best decision when available.

20. child `AiRun` detail: inspect prompt messages, role model, generation params, context pack, provider result, and sanitized raw response excerpt on JSON failures.

21. post-run editor state: inspect `local postDraft.versions[]` and

`finalText.editorDecisionSnapshot` to see which version the human selected, which comments created new versions, and whether machine trace context was available when the final text was approved.

22. post-run quality check: for human-comment revisions, inspect

`postDraft.versions[].qualityCheck` and child `AiRun.requestPayload.draftRunStep = humanCommentRevisionQualityCheck` to confirm matched/missed comment intents, source-marker preservation, public-prose status, internal jargon leaks, review attempts, incident metadata, and nested `operationEnvelope` on the revision/quality attempts.

23. post-run learning note: inspect `authorNotes[]` for type `editorialLearning`.

Pending and rejected notes should be visible but not used as author-position evidence. Accepted notes should flow through normal author-memory event and inference logic.

24. incident blast radius: when the same `incidentType` repeats across a run, compare

it against `CURRENT_LLM_OPERATION_INVENTORY` to identify other migrated or allowlisted operations with the same expected incident coverage before deciding whether this is an isolated provider error or a systemic architecture issue.

25. payload budget blast radius: when `qualityRisk`, `contextOverBudget`, or

`payloadTooLarge` repeats, compare `payloadBudget.profileId`, sent/trimmed counts, and inventory `payloadBudgetStatus` before blaming prompt quality.

10. Known AS IS limitations

- Editorial critique is active and feeds one alternative-angle challenger plus revision goals, but it is not yet an autonomous multi-agent debate loop.

- `anotherAngle` is active as one challenger route only, not as an autonomous debate loop or general fallback.

- ArticleDossier is a local compact memory artifact, not a vector store or cross-run RAG system.
- Failed late provider operations are safely finalized when the Python call returns or raises. Evidence interpretation now has an operation-level timeout envelope; a broader infrastructure watchdog is still needed for externally stuck Celery tasks outside protected operation envelopes.
- The pipeline has detailed traceability. The main editor UI receives one machine draft first, then manages post-run human versions locally. Cross-post learning is currently limited to reviewable editorialLearning author-memory notes; promotion into actual editorial rules remains a future slice.

11. Maintenance rules

- This Markdown document and the PDF must be updated together.
- Regenerate the PDF after every edit:

```
python scripts/generate-draft-run-pipeline-pdf.py
```

- Future DraftRun planning, implementation, diagnostics, evaluation, and autofix work must read this document before changing or judging the pipeline.
- If current behavior differs from this document, update the document first or include the update in the same slice as the behavior change.